When Nothing Changes, Who Changes? AI Exposure and Early-Career Employment in UK Occupations

Extended Essay

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Abstract

Current aggregate studies of AI often find small occupation-level employment effects. I argue that these results can mask within-occupation compositional change: AI reduces demand for junior workers while raising the marginal product of senior tasks. I formalize this mechanism by extending the expertise framework of Autor and Thompson (2025) to allow junior–senior task specialization. Using UK position data from Revelio Labs matched to LLM exposure scores, and treating the release of ChatGPT in November 2022 as a shock, I find that higher-exposure occupations experience a sustained decline in the junior hiring share, reaching 11.5 percentage points by 2025. Mobility evidence points to increased senior reallocation, which does not by itself identify net augmentation, and to larger seniority gains among transitioning juniors. I find no corresponding wage effects in predicted salary data.

1 Introduction

Since the launch of ChatGPT in autumn 2022, generative AI has developed quickly across text, code, image, and other creative tasks. This has raised a clear labour-market question: why have such large technical advances not yet produced large aggregate employment effects? Earlier computerisation produced visible structural change, including the occupational polarisation documented by Autor & Dorn (2013). One might therefore expect large language models to generate similarly visible disruption, especially because Eloundou et al. (2024) estimate that a large share of workers hold occupations with tasks exposed to LLMs.

Yet recent evidence is more muted at the aggregate level. The task framework of Acemoglu & Restrepo (2019) predicts that automation can both displace tasks and create new demand, so the net employment effect is ambiguous. Consistent with this, Massenkoff & McCrory (2026) find no systematic unemployment increase in highly exposed occupations, while Brynjolfsson et al. (2025) find sharper losses for early-career workers rather than broad occupational collapse. This suggests that the main adjustment may be hidden inside occupations.

I argue that aggregate null results can mask offsetting compositional shifts within occupations. AI may displace junior workers, whose tasks overlap more with current AI capabilities, while augmenting senior workers, whose expertise-intensive tasks remain complementary to AI. Since both groups sit inside the same occupation codes, junior displacement and senior augmentation may partly cancel in occupation-level employment measures. This idea is consistent with the seniority-biased effects documented by Lichtinger & Hosseini Maasoum (2025) and with evidence that early-career workers are especially exposed (Brynjolfsson et al., 2025).

This framing also helps connect two strands of evidence that otherwise look separate. Studies of aggregate occupation-level employment often find small or delayed effects,

while firm- or worker-level studies increasingly point to pressure on entry-level work. The contribution of this essay is to treat these as related facts rather than competing results: the same occupation can experience junior displacement and senior reallocation at the same time.

The theoretical contribution is to extend the expertise framework of Autor & Thompson (2025) by relaxing within-occupation worker homogeneity. In my partial-equilibrium CES model, junior and senior workers have comparative advantage in different tasks. AI substitutes for junior labour in low-expertise tasks and complements senior labour through task-level production. The model predicts a falling junior employment share, rising senior marginal products, ambiguous net occupation-level employment, and wage movements that may be hard to detect if quantity adjustment is stronger than price adjustment.

The empirical contribution is to test this within-occupation mechanism directly, rather than treating an occupation as a single homogeneous labour input. This is where individual-level position histories are especially useful for the research question studied here.

I test these predictions using UK position data from Revelio Labs matched to the LLM exposure scores of Eloundou et al. (2024). The empirical design treats the release of ChatGPT in November 2022 as a shock and uses cross-occupation variation in exposure. I find a sustained decline in the junior share of new positions in highly exposed occupations. I also study career mobility as a related adjustment margin, addressing the call by Brynjolfsson et al. (2025) and Klein Teeselink (2025) for more worker-level evidence. Senior workers in exposed occupations become more mobile, while junior workers who transition experience larger seniority gains; this estimate is stable after Heckman correction. I find no clear wage effects in the Revelio predicted salary data and no evidence of occupation switching away from exposed occupations.

The remainder of this essay proceeds as follows. Section 2 develops the theoretical framework. Section 3 describes the data and identification strategy. Section 4 presents the empirical results. Section 5 discusses limitations, and Section 6 concludes.

2 Theoretical Framework

Autor & Thompson (2025) introduce an expertise framework in which the effect of automation depends on which tasks are automated. When automation replaces inexpert tasks, the remaining task bundle becomes more expertise-intensive; wages may rise, but employment can fall because fewer workers meet the threshold. Their framework assumes that workers inside an occupation are homogeneous: each worker must perform all non-automated tasks and, conditional on meeting the expertise threshold, is equally productive.

This assumption is useful for their between-occupation analysis, but it rules out the within-occupation margin studied here. I keep the idea that occupations bundle tasks of different expertise levels, but allow junior and senior workers to specialise inside the same occupation. The motivation comes from knowledge-hierarchy models such as Garicano & Rossi-Hansberg (2006), where less experienced workers handle more routine problems and experienced workers solve harder ones. This gives a production-side reason why the same occupation code can contain different task portfolios. I use Revelio seniority as the empirical proxy for this distinction.

The model is deliberately partial equilibrium. I do not model how the whole economy reallocates workers across occupations, as in Autor & Thompson (2025). Instead, I ask what happens inside a given occupation when AI becomes more capable. This narrower setup is useful because the empirical analysis also works at the occupation-by-time and worker-transition levels. It allows the model to focus on the junior–senior composition margin without needing to explain all aggregate employment changes.

2.1 Model Setup

Consider a single occupation in partial equilibrium. Output Y is produced by combining three task types via CES aggregation:

$$Y = \left[\beta_L y_L^{\frac{\sigma-1}{\sigma}} + \beta_H y_H^{\frac{\sigma-1}{\sigma}} + \beta_G y_G^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \tag{1}$$

where $0 < \sigma < 1$ is the within-occupation task substitution elasticity and $\beta_L, \beta_H, \beta_G > 0$ are task share parameters, with $\beta_L + \beta_H + \beta_G = 1$. The parameter σ is distinct from the between-occupation elasticity λ in Autor & Thompson (2025).

Following Autor & Thompson (2025), tasks fall into three categories:

- Task L: low-expertise activities, including routine operations that are readily automatable by AI.
- Task H: high-expertise activities, including context-dependent judgement, managerial decision-making, and other functions requiring accumulated experience.
- Task G: generic support tasks assigned to junior workers, including common execution-oriented activities that remain non-automated.

The generic task allows junior workers to remain useful even when AI substitutes for part of their task bundle. This matters for generative AI because exposure is high in many white-collar occupations (Webb, 2020), where junior workers often perform both routine cognitive tasks and support work. Without task G, the model would predict too stark a form of displacement; with it, the junior share can fall without junior employment disappearing entirely.

Each task type is produced as follows:

$$y_L = A \cdot M + a_L \cdot n_J^L, \quad y_H = b_H \cdot n_S, \quad y_G = c \cdot n_J^G \tag{2}$$

In task L, AI capital M (rented at rate r) and junior labour n_J^L are perfect substitutes, with A denoting AI capability or cost-effectiveness in this task. This is the displacement channel. Task H requires senior workers because it involves judgement and accumulated experience. This creates the augmentation channel: AI raises the value of complemen-

tary high-expertise tasks. Task G is junior-assigned support work; I abstract from senior production of G because senior time has a higher return in task H.[1]

Junior workers allocate their labour between tasks L and G subject to $n_J^L + n_J^G = n_J$. At the occupation level, labour supply responds to wages with constant elasticity:

$$n_J = \varphi_J w_J^{\varepsilon}, \quad n_S = \varphi_S w_S^{\varepsilon} \tag{3}$$

where $\varepsilon > 0$ captures workers' ability to enter or exit the occupation in response to wage changes.

2.2 Equilibrium and Comparative Statics

The competitive firm's first-order conditions, under an interior solution in which juniors remain active in tasks L and G, yield three conditions; full derivations are in Appendix A.1 and Appendix A.3.

No-arbitrage in task L. Since AI and junior labour are perfect substitutes, the firm employs both only when their unit costs are equalised:

$$w_J = \frac{a_L \cdot r}{A} \tag{4}$$

Holding the AI rental rate r fixed, an increase in AI capability A lowers the junior wage w_J.

Internal allocation of junior labour. Equalizing the marginal revenue product of junior labour across tasks L and G yields $y_G/y_L = \gamma$, where $\gamma \equiv (\beta_G c/\beta_L a_L)^{\sigma}$ is independent of A.

Senior augmentation via complementarity. Senior wages satisfy:

$$w_S = p \cdot \beta_H \left(\frac{Y}{y_H}\right)^{\frac{1}{\sigma}} \cdot b_H \tag{5}$$

The term $(Y/y_H)^{1/\sigma}$ is the complementarity channel. Holding senior labour fixed, higher AI capability raises y_L and total output Y, which increases the marginal revenue product

[1]This restriction is equivalent to assuming that the senior return in task H exceeds the return seniors would obtain from generic support work. See Appendix A.2.

of task H and therefore w_S. The resulting rise in senior labour dampens but does not reverse this upward shift in senior demand.

Comparative statics. Locally around the interior solution, a rise in A lowers w_J and, with elastic labour supply, reduces n_J. Within junior labour, workers move away from task L toward task G, but total junior employment still falls. At the same time, the increase in y_L raises senior marginal products through CES complementarity, increasing w_S and n_S. The combined effect is a lower junior employment share. The net effect on total occupation-level employment remains ambiguous because the junior decline and senior increase work in opposite directions.

2.3 Testable Predictions

The static interior model formally delivers four implications:

1. **Compositional shift:** In occupations with higher AI exposure, the junior employment share declines following the AI shock, as junior workers are displaced from task L and exit the occupation.
2. **Senior augmentation:** Senior workers in high-exposure occupations experience rising marginal products, wages, and labour demand through CES complementarity.
3. **Wage adjustment:** Junior wages decline and senior wages rise. If the occupation-level supply elasticity ε is large, wage changes may be small relative to employment changes, but the model still predicts price adjustment.
4. **Aggregate ambiguity:** The net occupation-level employment effect is indeterminate in sign, consistent with the aggregate null results documented in the existing literature.

Because junior workers retain a role in task G, the model predicts partial rather than total displacement. The model is static and does not include search, matching, or skill accumulation, so the career-mobility results below should be read as complementary evidence rather than direct model tests.

3 Data and Identification Strategy

3.1 Data and Specification

I use individual-level position data from Revelio Labs, which aggregates UK LinkedIn profiles. The sample contains 23,305,509 workers and 50,742,348 position records active at some point between January 2019 and December 2025. For each position, I observe the O*NET occupation code, start and end dates, predicted salary, company id, seniority level, and within-individual position number. Table 1 shows the seniority scale.

Seniority	Level	Examples
1	Entry	Accounting Intern, Software Engineer Trainee, Paralegal
2	Junior	Account Receivable Bookkeeper, Junior Software QA Engineer, Legal Adviser
3	Associate	Senior Tax Accountant, Lead Electrical Engineer, Attorney
4	Manager	Account Manager, Superintendent Engineer, Lead Lawyer
5	Director	Chief of Accountants, VP Network Engineering, Head of Legal
6	Executive	Managing Director (Treasury), Director of Engineering (Backend Systems), Attorney (Partner)
7	Senior Executive	CFO, COO, CEO

Table 1: Revelio Labs seniority classification

I define a worker as *junior* if the position has seniority level 1 or 2.[2] This corresponds to Entry and Junior roles and captures 58.9 per cent of the sample. These workers typically perform execution-oriented tasks under supervision. From the Associate level onward, workers have more discretion and are closer to the senior task type in the model. Lichtinger & Hosseini Maasoum (2025) validate this measure by showing that job-title keywords align with expected seniority differences and remain stable around the diffusion of generative AI.

[2]The methodology of how Revelio Labs classify seniority is available at: https://www.data-dictionary.reveliolabs.com/methodology.html#seniority.

To measure AI exposure, I use the LLM exposure score of Eloundou et al. (2024), which estimates the share of task time that could be reduced by at least 50 per cent with access to a large language model. I merge this occupation-level score to Revelio using O*NET–SOC codes, with an 89.5 per cent match rate. The score enters the main regressions continuously. In the extensive-margin mobility sample, it ranges from 0 to 1, with mean 0.494, standard deviation 0.177, and interquartile range 0.147. I therefore interpret mobility coefficients using one-standard-deviation changes as well as full-scale changes. For robustness, I also use the AI Occupational Exposure index of Felten et al. (2021) and the AI Applicability Score of Tomlinson et al. (2025), reported in Appendix B.

The main compositional-shift analysis uses position starts from 2019H1 to 2025H2. I collapse the data into occupation × half-year cells and estimate

$$\text{JuniorShare}_{o,t} = \sum_{k \neq k_0} \beta_k \big(m_{1,o} \times \mathbb{1}[t = k] \big) + \alpha_o + \gamma_t + \varepsilon_{o,t} \tag{6}$$

where $\text{JuniorShare}_{o,t}$ is the share of new hires classified as junior in occupation o and half-year t; $m_{1,o}$ is LLM exposure; α_o and γ_t are occupation and half-year fixed effects; and 2022H2 is the omitted baseline. I use 2022H2 as the reference period because ChatGPT was released late in that half-year, leaving little time for hiring responses to appear in position start dates. I weight cells by the number of positions and cluster standard errors at the O*NET level. I also report a pooled difference-in-differences version with $m_{1,o} \times \text{Post}_t$, where Post_t begins in 2023H1.

For mobility, I construct consecutive position pairs for each worker and define the transition date as the start date of the next position. After cleaning, the dataset contains 12.6 million valid transitions.[3] The mobility post period begins in November 2022.

I separate the mobility analysis into an extensive margin and an intensive margin. The extensive margin asks whether a position is followed by another observed position

[3] I exclude individuals with more than 50 recorded positions (likely data artefacts), drop all positions for individuals with any overlapping employment spells to avoid ambiguous transition timing, and remove positions with missing start dates.

in the panel. The intensive margin conditions on a transition and asks what kind of move occurs. This distinction matters because the model predicts changes in the relative value of junior and senior work, but it does not itself contain a search or matching process.

The extensive margin specification estimates a linear probability model of the form

$$\text{Transition}_i = \delta_1 \text{AI}_o \times \text{Post}_i + \delta_2 \text{Junior}_i \times \text{AI}_o \times \text{Post}_i + \boldsymbol{X}_i' \eta + \alpha_o + \gamma_{s(i)} + \varepsilon_i \quad (7)$$

where Transition_i equals one if origin-position observation i is followed by another position; $\boldsymbol{X}_i$ includes lower-order interactions; and $s(i)$ indexes the origin start month. The coefficient δ_1 captures the senior AI-exposure effect after ChatGPT, while δ_2 captures the additional junior effect. Conditional on transition, I estimate seniority gains using

$$\Delta\text{Seniority}_i = \theta_1 \text{AI}_o \times \text{Post}_i + \theta_2 \text{Junior}_i \times \text{AI}_o \times \text{Post}_i + \boldsymbol{X}_i' \mu + \alpha_o + \gamma_{\tau(i)} + u_i \quad (8)$$

where $\tau(i)$ is the transition month. I also use log salary change and occupation switching as intensive-margin outcomes.

All mobility specifications include occupation and month fixed effects. The occupation fixed effects absorb permanent differences across occupations, including the level of AI exposure. The month fixed effects absorb common time shocks. Identification therefore comes from whether junior and senior workers in more exposed occupations change differentially after ChatGPT.

Because the intensive-margin regressions condition on transition, I also report a Heckman (1979) two-step correction. The concern is that the set of workers who transition after ChatGPT may not be random. For example, if weaker juniors are displaced and only stronger juniors move successfully, the observed seniority gain could overstate true upgrading. The selection equation uses the main regressors, occupation fixed effects, and a firm-level exclusion restriction: the pre-period mean separation rate at the origin firm, entered directly and interacted with junior status. The logic is that firms with historically high turnover create more transitions, but conditional on moving, their historical separation rate should not directly determine the seniority gain

in the next job. I estimate the probit on a 10 percent user-id subsample and include the inverse Mills ratio in the outcome equation. The corrected standard errors do not include Murphy–Topel or bootstrap adjustment, so they should be read cautiously.

3.2 Identification Strategy

The causal interpretation rests on two conditions. First, exposure must not be shaped by post-treatment labour-market adjustment. This holds by construction for the LLM exposure score, which is based on pre-existing O*NET task descriptions rather than firm adoption decisions. This differs from firm-level AI adoption measures, which may respond to workforce composition.

Second, the timing of the shock should be plausibly unanticipated. Figure 1 shows that public attention to "ChatGPT" was essentially zero before its release on 30 November 2022 and rose sharply afterward. I therefore use 2022H2 as the omitted period in Equation 6 and define the compositional-shift post period from 2023H1 onward.

Figure 1: Google Trends index for the keyword "ChatGPT", 2019–2025

The key identifying assumption is parallel evolution of junior hiring shares across exposure levels absent generative AI. I assess this using the pre-treatment event-study coefficients and a formal linear pre-trend test following Freyaldenhoven et al. (2019).

Two limitations remain. I cannot include firm × time fixed effects because treatment varies at the occupation level. Therefore, firm-level shocks correlated with occupation exposure cannot be fully ruled out. I also hold the junior–senior threshold fixed, although AI may change expertise requirements within occupations over time. Finally, the COVID period remains in the pre-period. Half-year and month fixed effects absorb common pandemic shocks, and the event study checks whether exposed occupations were already diverging before ChatGPT.

4 Results

I first document the main compositional result: the junior share of new hires falls in more AI-exposed occupations after ChatGPT. I then examine mobility outcomes to understand how workers adjust after the shock.

This result is related to the firm-level evidence in Klein Teeselink (2025), but the empirical margin is different. That study focuses on firm-level employment outcomes, while my main outcome is the junior hiring share within occupation–half-year cells. The estimates are therefore not directly comparable, but both point to junior workers bearing an important part of the early adjustment.

4.1 Compositional Shift

Figure 2 plots the monthly junior share of new position starts for occupations above and below the median AI exposure score. Before late 2022, both series are stable, with the high-exposure group around 55 per cent and the low-exposure group around 85 per cent. From early 2023, the high-exposure series falls sharply, dropping below 40 per cent by late 2025, while the low-exposure series declines only modestly. The timing is closely aligned with the diffusion of generative AI.

Figure 2: Monthly junior share of new position starts by AI exposure group. Junior defined as seniority ≤ 2 (Revelio Labs classification). AI exposure median split at 0.366 (Eloundou et al. (2024)). Vertical dashed line marks ChatGPT release (November 2022).

The pooled difference-in-differences estimate in Table 2 is -0.096. Thus, after ChatGPT, occupations with higher exposure experience a larger decline in the junior hiring share. A full-scale increase in exposure corresponds to a 9.6 percentage-point reduction.

	Coefficient	**Std. error**	t	**95% CI**
$\text{AI}_o \times \text{Post}_t$	-0.096	0.011	-8.71	$[-0.117, -0.074]$

Table 2: Pooled difference-in-differences estimate. Dependent variable: junior hiring share at the occupation–half-year level, weighted by cell size. Occupation and half-year fixed effects absorbed. Standard errors clustered at the O*NET code level. Estimation window: 2019H1–2025H2. $N = 12,760$.

The event study in Figure 3 provides the main identification check. The pre-treatment coefficients fluctuate around zero with no clear trend, and the 2022H1 coefficient is small and insignificant (-0.005, $p = 0.44$). There is a mild positive hump during 2019H2–2020H2, plausibly related to COVID-era differences across white-collar occupations, but it disappears before the treatment window. A formal linear pre-trend test following Freyaldenhoven et al. (2019) also gives an essentially zero slope. After 2023H1, the coefficients turn negative and grow in magnitude: -0.064 in 2023H1, -0.103 in 2024H1,

and -0.115 by 2025H2. All post-period estimates are significant at the 1 per cent level. The post-treatment decline is larger and more persistent than the pre-period fluctuations. The quarterly event study in Appendix Figure 6 shows a similar timing pattern.

Figure 3: Event study: effect of AI exposure on junior share of new position starts. Each point represents the estimated coefficient $\hat{\beta}_k$ from Equation 6. Estimation window: 2019H1–2025H2. Baseline period: 2022H2. Standard errors clustered at the O*NET code level; shaded area shows 95% confidence interval.

Overall, the evidence supports the first model prediction: high-exposure occupations experience a sustained decline in the junior share of new hires after the diffusion of generative AI.

The magnitude is economically meaningful. By 2025H2, the event-study estimate implies an 11.5 percentage-point lower junior hiring share for a fully exposed occupation relative to the baseline period. Even though the full zero-to-one exposure change is large, the direction and timing are clear: the break occurs after ChatGPT becomes widely known and continues through the post period.

The event-study design is important here because the pooled estimate alone would be hard to interpret if high- and low-exposure occupations were already on different paths. The absence of a clear pre-trend does not prove the identifying assumption, but it

makes the post-period break more credible. It also helps separate the AI-related decline from the temporary COVID-era fluctuations visible earlier in the sample.

4.1.1 Robustness Check

Appendix Appendix B reports robustness checks. The result is similar using the AI Occupational Exposure index of Felten et al. (2021) and under a binary exposure split. The AI Applicability Score of Tomlinson et al. (2025) gives a smaller and insignificant estimate, which is less surprising because it is based on observed usage rather than pre-determined task exposure. This measure may capture adoption frictions as well as exposure. Varying the junior cutoff shows that the effect is strongest at the baseline definition, seniority ≤ 2, and weaker when junior is defined only as level 1 or expanded to level 3. I therefore interpret the baseline result as evidence about workers below the expertise threshold, not about one exact job-title category.

4.2 Career Mobility

I next examine career mobility. These estimates are not direct tests of the static model, but they show how workers adjust after the compositional shift. I study transition probability, seniority gains, log salary changes, and occupation switching.

This worker-level analysis is useful for two reasons. First, a fall in the junior hiring share does not show what happens to workers who are already in exposed occupations. Second, gross flows may reveal reallocation even when net occupation-level employment changes are small. The mobility results should therefore be read as descriptive evidence on adjustment margins rather than as structural estimates of the model.

4.2.1 Extensive Margin: Who Moves?

Table 3 reports the extensive-margin results. The coefficient on AI $\times$ Post is 0.548, implying a 9.7 percentage-point increase in senior transition probability for a one-standard-deviation increase in exposure. This pattern is consistent with higher outside options and poaching for senior workers, but it could also reflect exits from worse

matches. I therefore interpret it as increased gross senior reallocation, not as direct evidence on net senior employment. The triple interaction is −0.660, so the AI-specific mobility gradient is weaker for juniors. Combining the two treatment terms gives about −0.11 per full-scale unit, or −2.0 percentage points for a one-standard-deviation increase. This is a relative AI-exposure effect, not an absolute decline in junior mobility, because the common post-period movement and the junior-specific post effect are also part of the total transition probability. The result is almost unchanged when excluding recent position starts (−0.662), suggesting that short-run right-censoring is not driving the estimate.

This pattern is consistent with the model's junior-displacement logic, but the senior-side interpretation is more cautious. The model predicts higher senior labour demand, whereas the empirical outcome here is a gross transition probability. A senior transition can be a move into a better match, but it can also be an exit from a weaker match. For this reason, I treat the extensive-margin result as evidence of reallocation rather than as a direct test of senior employment growth.

	Full Sample	**Excluding Recent**
Junior	–0.075***	–0.044***
	(0.010)	(0.010)
Junior × AI	0.205***	0.151***
	(0.019)	(0.018)
Junior × Post	0.377***	0.370***
	(0.009)	(0.008)
AI × Post	0.548***	0.567***
	(0.028)	(0.028)
Junior × AI × Post	–0.660***	–0.662***
	(0.031)	(0.029)
N	27,471,965	24,531,422
R^2	0.181	0.158
Occupation FE	Yes	Yes
Month FE	Yes	Yes

Table 3: Career mobility regressions: extensive margin (probability of job transition)

Notes: *Dependent variable is an indicator for whether the worker transitions to a new position. Column (1) uses the full sample; column (2) excludes positions starting after 2024m6 to assess censoring of recently started positions. AI exposure enters as the raw*

*LLM exposure score. In the full-sample estimation sample, its range is 0–1, its standard deviation is 0.177, and its interquartile range is 0.147. All specifications are linear probability models with occupation and month fixed effects. Standard errors clustered at the O*NET occupation level in parentheses.* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.2.2 Intensive Margin: Transition Outcomes

Conditional on transition, Table 4 shows larger seniority gains for juniors leaving high-exposure occupations after ChatGPT. The baseline triple interaction is 0.122, meaning that transitioning juniors in more exposed occupations move to higher-seniority roles relative to the comparison groups. The firm separation rate is a strong first-stage predictor of transition, and after Heckman correction the coefficient remains 0.125. The inverse Mills ratio is negative (−0.129), which suggests negative selection on the outcome dimension: workers more likely to transition tend to have smaller seniority gains. The stability of the treatment coefficient is therefore more important than the exact corrected standard error, and it points toward genuine skill-upgrading among transitioning juniors.

The salary and occupation-switching results are weaker. The triple interaction for log salary change is small and insignificant (0.007), even after controlling for seniority gains. The occupation-switching interaction is also small and insignificant (−0.008). Thus, the post-ChatGPT adjustment appears mainly in hiring composition and seniority progression rather than in wage compression or occupational flight.

The occupation-switching null is useful because it weighs against a simple story in which junior workers respond to AI exposure by leaving exposed occupations entirely. Instead, workers who move appear to remain in related occupational space while changing seniority level. This fits the idea that the early adjustment is within occupation, not only between occupations.

The wage null should also be read carefully. The theoretical model predicts wage movements, but Revelio salaries are predicted rather than directly observed compensation. It is therefore possible that true price effects are noisy or delayed in this dataset. The stronger evidence in this paper is on the quantity and composition margins.

	Seniority Gain		Log Salary Gain		Occupation Switch	
	(1)	(2)	(3)	(4)	(5)	(6)
Junior	1.534***	1.517***	–0.085***	–0.084***	0.059**	0.067**
	(0.049)	(0.049)	(0.014)	(0.014)	(0.027)	(0.027)
Junior × AI	–0.617***	–0.605***	0.031	0.030	–0.034	–0.040
	(0.094)	(0.094)	(0.026)	(0.026)	(0.047)	(0.047)
Junior × Post	–0.114***	–0.117***	0.032***	0.032***	–0.006	–0.004
	(0.015)	(0.015)	(0.007)	(0.007)	(0.007)	(0.007)
AI × Post	0.019	0.035	–0.111***	–0.112***	0.020	0.013
	(0.025)	(0.025)	(0.015)	(0.015)	(0.014)	(0.014)
Junior × AI × Post	0.122***	0.125***	0.007	0.006	–0.008	–0.009
	(0.030)	(0.030)	(0.013)	(0.013)	(0.013)	(0.013)
Δ Seniority			0.203***	0.203***		
			(0.002)	(0.002)		
IMR		–0.129***		0.011***		0.057***
		(0.014)		(0.003)		(0.006)
N	12,487,780	12,487,780	10,676,852	10,676,852	12,487,780	12,487,780
R^2	0.160	0.161	0.350	0.351	0.120	0.121
Occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Heckman correction	No	Yes	No	Yes	No	Yes

Table 4: Career mobility regressions: intensive margin outcomes conditional on job transition

Notes: *Sample restricted to workers observed in at least two positions. Columns (1)–(2) report seniority gain (levels); columns (3)–(4) report log salary change controlling for seniority gain; columns (5)–(6) report occupation switching (linear probability model). Odd columns are baseline OLS; even columns include Heckman selection correction using firm-level pre-period separation rate as the exclusion restriction. Standard errors clustered at the O*NET occupation level in parentheses. The standard errors in the Heckman columns do not incorporate Murphy–Topel or bootstrap correction for generated-regressor uncertainty and may be understated. Significance stars in those columns should therefore be interpreted cautiously.* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5 Discussion

A natural question is why I do not detect the wage movements predicted by the model. In the theory, junior employment falls because w_J falls, while senior employment rises because w_S rises. A high labour supply elasticity ε can make wage changes small relative to quantity changes, but it does not remove the price response altogether. The wage null should therefore be read as a limitation of the empirical test rather than as confirmation of the wage predictions. One important caveat is that Revelio salaries are

predicted from job titles, locations, and company characteristics, rather than directly observed compensation. This means the wage evidence is available, but less reliable than the employment and mobility evidence. Better administrative wage data would help distinguish true wage nulls from delayed effects or measurement smoothing.

The Heckman correction gives a related lesson. The inverse Mills ratio is negative ($\hat{\lambda} = -0.129$), suggesting negative selection on the outcome dimension: workers more likely to transition tend to have smaller seniority gains. The key point, however, is that the triple-interaction estimate is almost unchanged after correction (0.125 versus 0.122). Since the corrected standard errors do not include Murphy–Topel or bootstrap adjustment, I place more weight on the stability of the point estimate than on its exact precision. The preferred interpretation is genuine skill-upgrading among transitioning juniors, although this remains a career-progression finding beyond the static model. The occupation-switching null supports this reading: juniors in high-exposure occupations are not mainly fleeing into unrelated occupations, but moving upward within related occupational space.

The extensive-margin result should also be interpreted cautiously. Senior workers in high-AI-exposure occupations become more mobile after ChatGPT, while the AI-specific mobility gradient is weaker for juniors. This does not mean that junior transition probabilities fall in absolute terms: the coefficient $\beta_4 + \beta_5$ only captures how the post-period mobility effect changes with AI exposure. More importantly, the senior result is a gross transition measure, not a net stock measure. A senior transition may reflect poaching into a better match, which is consistent with augmentation, or exit from a worse match. Distinguishing these channels would require occupation-level inflows, outflows, and net senior headcount changes. Right-censoring also remains a limitation: excluding very recent position starts is useful, but a stronger test would impose a common follow-up window or estimate a duration model.

With richer linked employer–employee data, a further extension would be to use the theoretical model more quantitatively. Direct measures of task allocation, compensation, and occupation-level flows could help discipline the task substitution elasticity, labour supply elasticity, and task weights. This would make it possible to quantify the displacement and augmentation channels, rather than only signing them.

A broader concern is the lack of consensus on how to measure occupational exposure to AI. Gimbel et al. (2026) show that common indices disagree on which occupations are most exposed, partly because they define exposure through different objects: task descriptions, technical capabilities, patents, or observed use. This motivates my use of three measures: the primary LLM exposure score, the AI Occupational Exposure index, and the AI Applicability Score. The last should be interpreted cautiously because it is based on observed Copilot use during the post-treatment period. It is useful as a revealed-use comparison, but weaker as an exogenous exposure measure. A separate data limitation is that Revelio likely over-represents white-collar professional workers, where LLM exposure is especially concentrated. This may amplify the estimated effects relative to a nationally representative labour-force dataset.

Finally, the seniority-cutoff robustness checks suggest that the effect is strongest around the junior–senior boundary, not at the very lowest entry level. One possible interpretation is that AI exposure bites most sharply after entry-level training but before workers cross the expertise threshold into more autonomous tasks. Level-2 workers may perform the drafting, coordination, coding, summarisation, and analytical support tasks that current AI systems can substitute for, while level-1 roles may be partly protected by training pipelines, supervision requirements, or noisier job-title classification. This distinction between entry-level access and early-career progression is important for future work. More broadly, these estimates capture the early stages of generative AI diffusion. As AI capabilities move from text-based assistance toward more autonomous task execution, current occupational task bundles may not remain stable. Longer post-

treatment panels will be needed to show whether these compositional shifts persist, intensify, or reverse.

These limitations do not overturn the main result, but they define its scope. The paper is best read as evidence on early, within-occupation adjustment to generative AI, rather than as a complete account of long-run labour-market effects.

In this sense, the results are most useful as a warning about where adjustment may first appear. Aggregate occupation totals may remain stable even while entry-level pathways inside those occupations narrow.

6 Conclusion

This essay has argued that aggregate null results for AI's employment effects conceal meaningful compositional shifts within occupations. Extending the expertise framework of Autor & Thompson (2025) to permit within-occupation task specialisation, I derive formal predictions for how AI capability improvements affect junior employment shares, senior marginal products, wages, and net occupation-level employment. The evidence is strongest for the model's junior compositional prediction: in occupations more exposed to large language models, the junior hiring share declines. The worker-level mobility results provide complementary evidence on adjustment margins beyond the static model. Juniors who transition achieve larger seniority gains, consistent with genuine skill-upgrading, while senior workers become substantially more mobile. These results should be interpreted as career-progression and gross-reallocation patterns rather than direct tests of formal model predictions. By contrast, I do not detect the wage movements predicted by the formal model in the Revelio predicted salary data, so the price-margin results should be interpreted cautiously.

These findings carry a subtler implication. Much of the public debate frames AI as a threat to entire occupations or skill groups. The evidence here suggests a different pattern: AI reshapes *who does the work within an occupation* before it reshapes *whether*

the occupation exists. If this pattern generalises, the workers most immediately affected are those at the bottom of the ladder within occupations that may still be thriving overall. This is a form of displacement that aggregate statistics are poorly equipped to detect, and that conventional retraining policies may be poorly designed to address. The relevant policy question may therefore be how firms and institutions restructure entry-level pathways when the tasks that once served as training grounds are increasingly performed by machines.

Whether these early compositional shifts prove transitory or permanent, and whether the expertise threshold itself adapts in response, are questions that longer post-treatment data will help resolve. But if displacement does begin at the junior level, as the evidence here suggests, the implications extend beyond the immediately affected workers. Every senior professional was once a junior one; entry-level positions have traditionally served as the channel through which new labour market entrants acquire occupation-specific skills and progress toward independent judgment. A sustained contraction of this channel would warrant attention from policymakers and educational institutions, not because entire occupations are disappearing, but because the pathway into them is narrowing.

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Appendix

A Derivation of Theoretical Model

A.1 Firm's Optimisation Problem and First-Order Conditions

The representative firm in the occupation takes output price p and factor prices w_J, w_S, r as given, and chooses AI capital M, junior labour in task L (n_J^L), junior labour in task G (n_J^G), and senior labour n_S to maximise profit:

$$\max_{M, n_J^L, n_J^G, n_S} \quad p \cdot Y - w_J\left(n_J^L + n_J^G\right) - w_S n_S - rM \tag{9}$$

where output Y is the CES aggregate defined in Equation 1, and the three task production functions are given by Equation 2.

Substituting the CES structure, the marginal revenue product of task $k \in \{L, H, G\}$ is:

$$p \cdot \frac{\partial Y}{\partial y_k} = p \cdot \beta_k \left(\frac{Y}{y_k}\right)^{\frac{1}{\sigma}} \tag{10}$$

The first-order conditions for an interior solution are:

$$p \cdot \beta_L \left(\frac{Y}{y_L}\right)^{\frac{1}{\sigma}} \cdot A = r \tag{11}$$

$$p \cdot \beta_L \left(\frac{Y}{y_L}\right)^{\frac{1}{\sigma}} \cdot a_L = w_J \tag{12}$$

$$p \cdot \beta_G \left(\frac{Y}{y_G}\right)^{\frac{1}{\sigma}} \cdot c = w_J \tag{13}$$

$$p \cdot \beta_H \left(\frac{Y}{y_H}\right)^{\frac{1}{\sigma}} \cdot b_H = w_S \tag{14}$$

Derivation of the no-arbitrage condition. Dividing Equation 11 by Equation 12:

$$\frac{A}{a_L} = \frac{r}{w_J} \quad \Rightarrow \quad w_J = \frac{a_L \cdot r}{A} \tag{15}$$

Since AI and junior labour are perfect substitutes in task L, the firm employs both only when their unit costs are equalised. This condition pins junior wages entirely as a function of AI capability A and the rental rate r, independently of conditions in tasks H and G.

Derivation of the constant task output ratio. Dividing Equation 12 by Equation 13:

$$\frac{\beta_L (Y/y_L)^{\frac{1}{\sigma}} \cdot a_L}{\beta_G (Y/y_G)^{\frac{1}{\sigma}} \cdot c} = 1 \tag{16}$$

Simplifying:

$$\frac{\beta_L a_L}{\beta_G c} \cdot \left(\frac{y_G}{y_L}\right)^{\frac{1}{\sigma}} = 1 \quad \Rightarrow \quad \frac{y_G}{y_L} = \left(\frac{\beta_G c}{\beta_L a_L}\right)^{\sigma} \equiv \gamma \tag{17}$$

The ratio γ is a structural constant determined by technology and task share parameters. Intuitively, both tasks L and G are produced using the same marginal factor (junior labour at wage w_J), so their marginal costs move simultaneously. The CES demand system then fixes their output ratio regardless of A.

Senior wage expression. From Equation 14 directly:

$$w_S = p \cdot \beta_H \left(\frac{Y}{y_H}\right)^{\frac{1}{\sigma}} \cdot b_H \tag{18}$$

This is the marginal revenue product of senior labour in task H.

A.2 Senior Allocation to Task H

The main text writes task G as a junior-assigned generic task and therefore abstracts from senior production of G. This is a simplifying modelling restriction, not a separate empirical claim. It can be interpreted as the corner solution of a more general model in which senior workers could perform generic support tasks but face a higher opportunity cost of time.

If a senior worker were reallocated to task G, the return would be no higher than the marginal revenue product of junior workers in task G, which by Equation 13 is exactly w_J. The return to senior labour in task H is w_S. Senior allocation to task H is therefore consistent with optimisation whenever $w_S > w_J$, that is:

$$p \cdot \beta_H \left(\frac{Y}{y_H}\right)^{\frac{1}{\sigma}} \cdot b_H > \frac{a_L \cdot r}{A} \tag{19}$$

This condition holds when task H is sufficiently valued relative to routine and generic support tasks. This is empirically plausible: in the Revelio Labs data, salaries for senior positions (seniority ≥ 5) substantially exceed those for entry-level positions (seniority ≤ 2) within the same O*NET occupation. I maintain $w_S > w_J$ as a regularity condition throughout.

A.3 Comparative Statics

I derive the effects of an increase in AI capability A on the equilibrium allocation in four steps.

Step 1: Junior wage and employment

From the no-arbitrage condition:

$$\frac{\partial w_J}{\partial A} = -\frac{a_L r}{A^2} < 0 \tag{20}$$

Substituting into the supply function (Equation 3):

$$n_J = \varphi_J \left(\frac{a_L r}{A}\right)^{\varepsilon} \quad \Rightarrow \quad \frac{\partial n_J}{\partial A} = -\varepsilon \varphi_J (a_L r)^{\varepsilon} A^{-(\varepsilon+1)} < 0 \tag{21}$$

An increase in AI capability monotonically reduces both the wage and employment of junior workers in the occupation. The magnitude of the employment response is governed by the supply elasticity ε: when ε is large, even a small decline in w_J induces substantial exit. Pinning down the precise value of ε is beyond the scope of this essay; the key qualitative prediction — that junior employment falls with AI capability — holds for any $\varepsilon > 0$.

Step 2: Task output levels

To characterize how task outputs respond to A, I reduce the model's dimensionality. Since $y_G = \gamma y_L$ with γ constant, I define $\tilde{\beta}_L \equiv \beta_L + \beta_G \gamma^{\frac{\sigma-1}{\sigma}}$ and write the CES aggregate as a function of two effective task outputs:

$$Y = \left[\tilde{\beta}_L y_L^{\frac{\sigma-1}{\sigma}} + \beta_H y_H^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \tag{22}$$

In this reduced representation, increasing y_L along the optimal path also requires increasing y_G by γ units. The marginal value of the effective low-expertise bundle is therefore:

$$p \cdot \tilde{\beta}_L \left(\frac{Y}{y_L}\right)^{\frac{1}{\sigma}} \tag{23}$$

The corresponding unit cost is not just the direct AI cost r/A. One additional unit of y_L can be produced by AI at cost r/A, but maintaining $y_G = \gamma y_L$ also requires γ/c units of junior labour in task G, paid $w_J = a_L r/A$. Hence the effective unit cost is:

$$q_L(A) \equiv \frac{r}{A} 1 + \frac{a_L \gamma}{c} \tag{24}$$

The reduced first-order condition is:

$$p \cdot \tilde{\beta}_L \left(\frac{Y}{y_L}\right)^{\frac{1}{\sigma}} = q_L(A) \tag{25}$$

This condition is consistent with the original three-task FOCs. Since γ is defined by Equation 12 and Equation 13,

$$\beta_G \gamma^{\frac{\sigma-1}{\sigma}} = \beta_L \frac{a_L \gamma}{c}, \quad \tilde{\beta}_L = \beta_L 1 + \frac{a_L \gamma}{c} \tag{26}$$

so Equation 25 is exactly Equation 11 scaled up by the cost and value of the induced increase in task G. Holding y_H fixed, an increase in A lowers $q_L(A)$. Since CES marginal value is decreasing in y_L for fixed y_H, the optimal y_L rises. Since $y_G = \gamma y_L$, task G output rises proportionally.

To verify that total output Y rises: from Equation 22, Y is increasing in both y_L and y_H. Since y_L increases and y_H is (initially) unchanged, Y unambiguously increases.

Step 3: Senior augmentation

I establish that the senior wage schedule shifts upward when A increases, and that the equilibrium senior wage and employment both rise.

Define the inverse demand for senior labour, holding A fixed:

$$w_S^D(n_S; A) \equiv p \cdot \beta_H \left(\frac{Y(n_S, A)}{b_H n_S} \right)^{\frac{1}{\sigma}} \cdot b_H \tag{27}$$

where $Y(n_S, A)$ is total output evaluated at the optimal task allocation for given n_S and A.

For the comparative static, the key property is the shift in this schedule rather than strict downward slope. Under the maintained regularity that w_S^D is weakly non-increasing in n_S for fixed A–with equality possible under constant returns–an upward shift of this schedule raises both the equilibrium wage and employment of senior labour.

Holding n_S fixed, an increase in A lowers the effective unit cost $q_L(A)$, raising y_L and $y_G = \gamma y_L$. Since $y_H = b_H n_S$ is unchanged, total output Y increases while y_H does not. Therefore Y/y_H rises and so does the senior marginal revenue product:

$$\frac{\partial w_S^D(n_S; A)}{\partial A} > 0 \quad \text{for fixed } n_S \tag{28}$$

The supply curve $n_S = \varphi_S w_S^{\varepsilon}$ is upward-sloping. The higher senior wage schedule therefore implies that the new interior equilibrium features both higher w_S and higher n_S:

$$\frac{\partial w_S}{\partial A} > 0, \quad \frac{\partial n_S}{\partial A} > 0 \tag{29}$$

The increase in n_S also raises $y_H = b_H n_S$ in equilibrium. This feedback lowers Y/y_H relative to the impact effect at fixed n_S, but it does not overturn the comparative static because the senior wage schedule has shifted upward. This confirms the augmentation result: AI capability improvements raise senior wages and senior employment via CES task complementarity.

Step 4: Compositional shift

Define the junior employment share as $s_J \equiv n_J/(n_J + n_S)$. From Steps 1 and 3, $\partial n_J/\partial A < 0$ and $\partial n_S/\partial A > 0$. For $n_J, n_S > 0$:

$$\frac{\partial s_J}{\partial A} = \frac{n_S \cdot \partial n_J/\partial A - n_J \cdot \partial n_S/\partial A}{(n_J + n_S)^2} < 0 \tag{30}$$

The junior employment share therefore unambiguously declines.

Internal reallocation of junior labour

Although total junior employment falls, the allocation of remaining juniors between tasks L and G also shifts. From $y_G = \gamma y_L$ and $y_G = c \cdot n_J^G$:

$$n_J^G = \frac{\gamma y_L}{c} \tag{31}$$

Since y_L is increasing in A (Step 2), n_J^G increases. Combined with declining total junior employment (n_J falls), this implies:

$$n_J^L = n_J - n_J^G \tag{32}$$

which falls faster than n_J itself. Junior workers are squeezed out of task L on two margins: AI directly substitutes their labour, and the remaining juniors reallocate toward task G where AI cannot substitute.

The above derivations assume an interior solution in which $n_J^L > 0$. For sufficiently large A, all junior workers exit task L and produce only task G output. I focus on the interior case, which is empirically relevant given that junior workers remain present in high-exposure occupations throughout the sample period.

A.4 Net Occupation-Level Employment

The net effect of AI capability on total occupation-level employment $N = n_J + n_S$ is:

$$\frac{\partial N}{\partial A} = \frac{\partial n_J}{\partial A} + \frac{\partial n_S}{\partial A} \tag{33}$$

The first term is unambiguously negative and the second is unambiguously positive (from my derivation in Appendix A.3). The net sign depends on the relative magnitudes, which are governed by the structural parameters. Senior augmentation dominates ($\partial N/\partial A > 0$) when:

- The task share of high-expertise tasks (β_H) is large, amplifying the complementarity channel.
- Senior supply is highly responsive (φ_S large or ε large), so that wage increases translate into large employment gains.

Conversely, junior displacement dominates ($\partial N/\partial A < 0$) when β_L is large, junior supply is highly responsive (φ_J large or ε large), and the initial junior share is high.

B Robustness Check

B.1 Summary of Alternative Specifications

Specification	**Treatment**	**Junior def.**	**Coefficient**	**Std. error**
Main	z(Eloundou)	≤ 2	-0.021^{***}	(0.002)
R1: Felten	z(Felten)	≤ 2	-0.018^{***}	(0.003)
R2: Tomlinson	z(Tomlinson)	≤ 2	-0.005	(0.003)
R3: Seniority 1	z(Eloundou)	≤ 1	-0.003	(0.003)
R4: Seniority 3	z(Eloundou)	≤ 3	-0.009^{***}	(0.003)
R5: Binary	Binary(Eloundou)	≤ 2	-0.039^{***}	(0.005)

Table 5: Robustness of the pooled difference-in-differences estimate for the junior hiring share. All continuous treatment variables are standardised to have zero mean and unit standard deviation at the occupation level. The dependent variable is the mean junior share within each occupation–half-year cell, weighted by cell size. Standard errors in parentheses, clustered at the O*NET code level. $***p < 0.01$.

Notes. The baseline period is 2022H2. "z(·)" denotes the occupation-level z-score of the indicated measure. R1 uses the AI Occupational Exposure index of Felten, Raj and Seamans (2021). R2 uses the AI Applicability Score of Tomlinson et al. (2025). R5 defines high exposure as above-median Eloundou et al. (2024) score.

C Other Statistics

Period	$\hat{\beta}_k$	95% CI lower	95% CI upper
2019H1	0.005	−0.011	0.020
2019H2	0.018	0.006	0.030
2020H1	0.012	−0.008	0.032
2020H2	0.024	0.008	0.039
2021H1	0.003	−0.013	0.018
2021H2	0.007	−0.001	0.015
2022H1	−0.005	−0.018	0.008
2022H2	**0**		
2023H1	−0.064	−0.090	−0.038
2023H2	−0.062	−0.082	−0.043
2024H1	−0.103	−0.136	−0.071
2024H2	−0.098	−0.124	−0.072
2025H1	−0.114	−0.151	−0.077
2025H2	−0.115	−0.150	−0.080

Table 6: Event study coefficients from Equation 6. The dependent variable is the junior hiring share at the occupation–half-year level, weighted by cell size. The estimation window is 2019H1–2025H2. The omitted baseline is 2022H2. Standard errors clustered at the O*NET code level (917 clusters).